Shlok Vaishnav

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Education

Institute of Technology, Nirma University Aug 2024 – May 2028
B.Tech in Computer Science and Engineering, CGPA: 7.58 *Ahmedabad, Gujarat, India*

- **Relevant Coursework:** Data Structures & Algorithms, Design and Analysis of Algorithms, Operating Systems, Computer Networks, Database Management Systems, Discrete Mathematics

Experience

Space Applications Centre, ISRO Feb 2026 – Present
Computer Vision Researcher *Ahmedabad, Gujarat, India*

- Achieved **0.83 mIoU** on lunar crater tiles by designing and training a two-stage crater detection and segmentation pipeline on Chandrayaan-2 terrain-model (DTM) imagery.

Bull Agritech Apr 2026 – Sep 2026
Software Engineering Intern *Ahmedabad, Gujarat, India*

- Built XGBoost price-forecasting models for non-perishable commodities across Gujarat's APMC markets, beating the seasonal-naive baseline by **40%** on held-out seasons.
- Developed procurement-timing signals from price z-scores and crop-arrival data across **20 collection centers** to identify optimal buying windows.
- Shipped a daily pipeline ingesting mandi arrivals and prices for **200+ commodity-market pairs**, and backtested weather-driven supply-shock indicators over **5 seasons** to surface procurement risk early.

Research Projects

Replica-Recall-Divergence: ANN Replication Under Chaos | *C++17, Raft, HNSW, gRPC, Python* GitHub

- Built a distributed vector database from scratch (Raft, HNSW, gRPC replication, consistent hashing) as a controllable testbed for silent search-quality loss in replicated approximate indexes, a failure mode Jepsen-style analyses had not targeted.
- Designed a chaos-injection protocol that probes each replica directly, bypassing coordinator scatter-gather, and showed via Mann–Whitney U tests (5 seeds, SIFT1M) that node failures cause significant, non-recovering recall divergence ($\mathbf{p = 0.0079}$) while healthy replicas agree to **1e-4**.

Null-Scaffold-Audit: Equivalence Testing for NAS Scaffolds | *Python, SciPy, Docker* GitHub

- Built a pre-registered TOST equivalence-testing audit (BCa bootstrap, Tango-score intervals, Holm correction) that decides before publication whether a scaffolded NAS pipeline beats budget-matched restarts of its own bare searcher.
- Validated the audit with a **165-test suite at 98.23% coverage**, which caught a fixed-seed bug in an in-house pipeline (identical output across all three "refine" iterations) before benchmarking NAS controllers on NATS-Bench.

Mini-vLLM: From-Scratch LLM Inference Engine | *Python, PyTorch, Triton, CUDA, FastAPI* GitHub

- Built an LLM inference engine from scratch with continuous batching and a paged KV-cache, verified token-for-token against Hugging Face transformers.
- Raised throughput from **40 to 211 tokens/sec** (**5.3**×) at P95 latency under **240ms** with a custom Triton paged-attention kernel, and cut model size up to **4**× with INT8/INT4 quantization.

Technical Skills

Languages: C++, Java, Python, SQL, JavaScript, TypeScript, Bash
Frameworks & Backend: Django, DRF, FastAPI, Next.js, React, gRPC
Databases, Infra & ML: PostgreSQL, Redis, Kafka, CUDA, PyTorch
Tools: Git, Docker, GCP, DVC, GitHub Actions, Prometheus, Grafana
Core Competencies: Software Design, Distributed Systems, Performance Optimization, Data Structures & Algorithms